Week 1

Date: Aug 2026

Project Goal

I wanted to understand how aircraft wings generate lift and why airplanes are able to stay in the air despite their weight.

What I Worked On

- Researched the basic principles of aerodynamics.
- Studied the lift equation and its variables.
- Learned about air density, velocity, wing area, and lift coefficient.
- Created the GitHub repository and project outline.
- Designed the project visual and README structure.

What I Learned

I learned that lift depends on several factors working together rather than a single force. Small changes in speed or wing characteristics can significantly affect aircraft performance.

Challenges

At first, I found it difficult to understand the lift coefficient and how engineers estimate its value in real aircraft.

Results

- Project structure completed.
- Research phase completed.
- First version of the lift equation implemented in Python.

Next Steps

- Complete the working calculator.
- Test different flight conditions.
- Verify calculations with sample values.

Week 2

Date: Aug 2026

What I Worked On

- Completed the first working version of the Aircraft Wing Lift Calculator.
- Added user inputs for:
    - Air Density
    - Velocity
    - Wing Area
    - Lift Coefficient
- Calculated lift force using the standard lift equation.
- Improved project documentation.
- Finalized project visuals for GitHub.

What I Learned

Working on this project helped me understand how engineers combine mathematics, physics, and programming to solve real-world problems.

I also learned that the same aircraft can generate very different amounts of lift depending on altitude, air density, and flight speed.

Results

The calculator successfully calculated lift force using different flight conditions and input values.

Example:

Air Density = 1.225 kg/m^3

Velocity = 120 m/s

Wing Area = 30 m^2

Lift Coefficient = 0.85

Output:

Lift Force ≈ 224,910 N

Challenges

One of the parts I struggled with at first was understanding how aerodynamic concepts are turned into mathematical equations that can be used in a computer program.

Key Lessons Learned

- Lift is generated through the interaction of multiple variables.
- Mathematics provides the foundation for engineering analysis.
- Programming can be used to model and visualize engineering concepts.
- Engineering is about understanding relationships between variables and predicting outcomes.

Final Reflection

This project helped me better understand one of the most fundamental questions in aviation:

How does an aircraft stay in the air?

Before starting this project, I understood the basic concept of lift, but I had never explored the factors behind it in detail. Building a simple calculator allowed me to connect physics, mathematics, and programming in a practical way.

The most rewarding part was seeing how a formula I had only read about could become a working tool that I built myself using Python.

Flight begins long before takeoff. It begins with understanding the science behind it.